

5G NR Frame Structure Analysis with Different Subcarrier Spacing

1. Aim

To analyze the 5G NR frame structure by studying the effect of different subcarrier spacings, slot durations, and OFDM symbols using MATLAB

2. Software Used

MATLAB

3. Code

```
clc;
clear;
close all;

% 5G NR Numerology
mu = 0:4;

% Subcarrier Spacing (kHz)
SCS = [15 30 60 120 240];

% Slot Duration (ms)
slotDuration = 1./(2.^mu);

% Display Table
disp('5G NR Frame Structure Analysis');
disp('-----');
disp(table(mu,SCS,slotDuration,...
    'VariableNames',{'Numerology','SCS_kHz','SlotDuration_ms'}));

% Plot 1: Subcarrier Spacing
figure;
subplot(2,1,1)
bar(mu,SCS)
grid on
xlabel('Numerology Index (\mu)')
ylabel('Subcarrier Spacing (kHz)')
title('5G NR Subcarrier Spacing')
```

```

% Subcarrier Spacing (kHz)
SCS = [15 30 60 120 240];

% Slot Duration (ms)
slotDuration = 1./(2.^mu);

% Display Table
disp('5G NR Frame Structure Analysis');
disp('-----');
disp(table(mu',SCS',slotDuration',...
    'VariableNames',{'Numerology','SCS_kHz','SlotDuration_ms'}));

% Plot 1: Subcarrier Spacing
figure;
subplot(2,1,1)
bar(mu,SCS)
grid on
xlabel('Numerology Index (\mu)')
ylabel('Subcarrier Spacing (kHz)')
title('5G NR Subcarrier Spacing')

% Plot 2: Slot Duration
subplot(2,1,2)
plot(mu,slotDuration,'-o','LineWidth',2)
grid on
xlabel('Numerology Index (\mu)')
ylabel('Slot Duration (ms)')
title('5G NR Slot Duration')

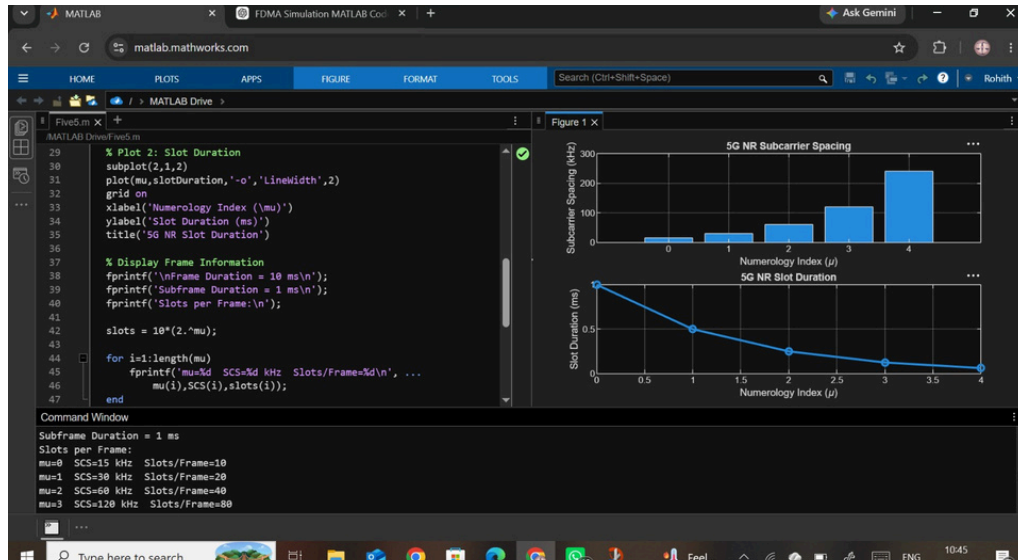
% Display Frame Information
fprintf('\nFrame Duration = 10 ms\n');
fprintf('Subframe Duration = 1 ms\n');
fprintf('Slots per Frame:\n');

slots = 10*(2.^mu);

for i=1:length(mu)
    fprintf('mu=%d SCS=%d kHz Slots/Frame=%d\n', ...
        mu(i),SCS(i),slots(i));
end

```

6. Output Graphs



7. Result

The 5G NR frame structure was successfully analyzed using MATLAB. The simulation shows that increasing the subcarrier spacing from 15 kHz to 240 kHz decreases the slot duration and increases the number of slots per frame. This demonstrates the flexible numerology of 5G NR, which supports different latency and bandwidth requirements.